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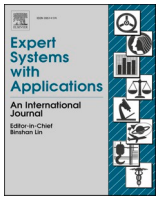

A visualization system for dam safety monitoring with application of digital twin platform

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ABSTRACT

For decades, research on dam safety monitoring systems has received considerable critical attention due to their significance in ensuring public safety and social economic development. In view of the rapid development of computational technologies, it's becoming increasingly difficult to ignore the urgent need to accelerate the intelligent development of dam safety monitoring systems. The emphasis of the prior studies usually lay in the dam safety monitoring data and structural health analysis. However, they neglect the importance of the real-time interactions of the systems, which makes the presented work not suitable for today's digital age. As such, based on digital twin platform, this paper aims to propose a visualization system for dam safety monitoring. Primarily, the next-generation and real-time rendering technologies are introduced to construct digital twin platform. The digital twin platform serves as a digital sand table to accurately reflect the dam current state and real-time monitoring information. Subsequently, artificial intelligence algorithms and numerical simulation are applied in monitoring data analysis, prediction model construction, structural safety evaluation, and risk early warning. Based on the windows platform, all of the above system functions are implemented in C++, Python languages, and OpenGL. Eventually, this presented real-time digital twin system was applied in a real dam in China. The direct outcome shows its superiority and effectiveness in a continuous and accurate dam safety evaluation.

1. Introduction

In today's society, dams have a significant impact on water supply, irrigation, flood control, and power generation. During their servicing period, a large number of dams are facing the challenges of unusual weather, aging, and increasing external load. Therefore, dam damage and failures sometimes may occur, which will lead to a number of casualties and enormous loss of property. To prevent the collapse of dams, developing a more scientific and intelligent dam safety monitoring system is a research hotspot in the field of dam disaster management.

During the past decades, the pioneer work can be traced to some dam-safety-related information systems overseas (Gorini et al., 2010). But these works have only considered the basic databases or materials, and ignored the automation and diverse functions of the systems (Man et al., 2018). The advent of computer technologies and new dam health diagnose theories accelerate the development of dam safety monitoring system. In the late 1980s, to analyse dam safety, a function to evaluate the flood hazard was applied in the system DAMSAFE (Bocchiola & Rosso, 2014), which helps to approximate but comprehensive assessment of flood hazard and dam risk. For the purpose of ensuring dam consistent safety, the Korea Water Resource Corporation (Kwater) has developed a dam safety management system, which is called KDSMS in 2009 (Jeon et al., 2009). Although a lot of effort has been spent on promoting KDSMS to serve as a multi-functional enterprise management system, including dealing with safety management of each field, research centre, headquarters office and their correlation, it can only

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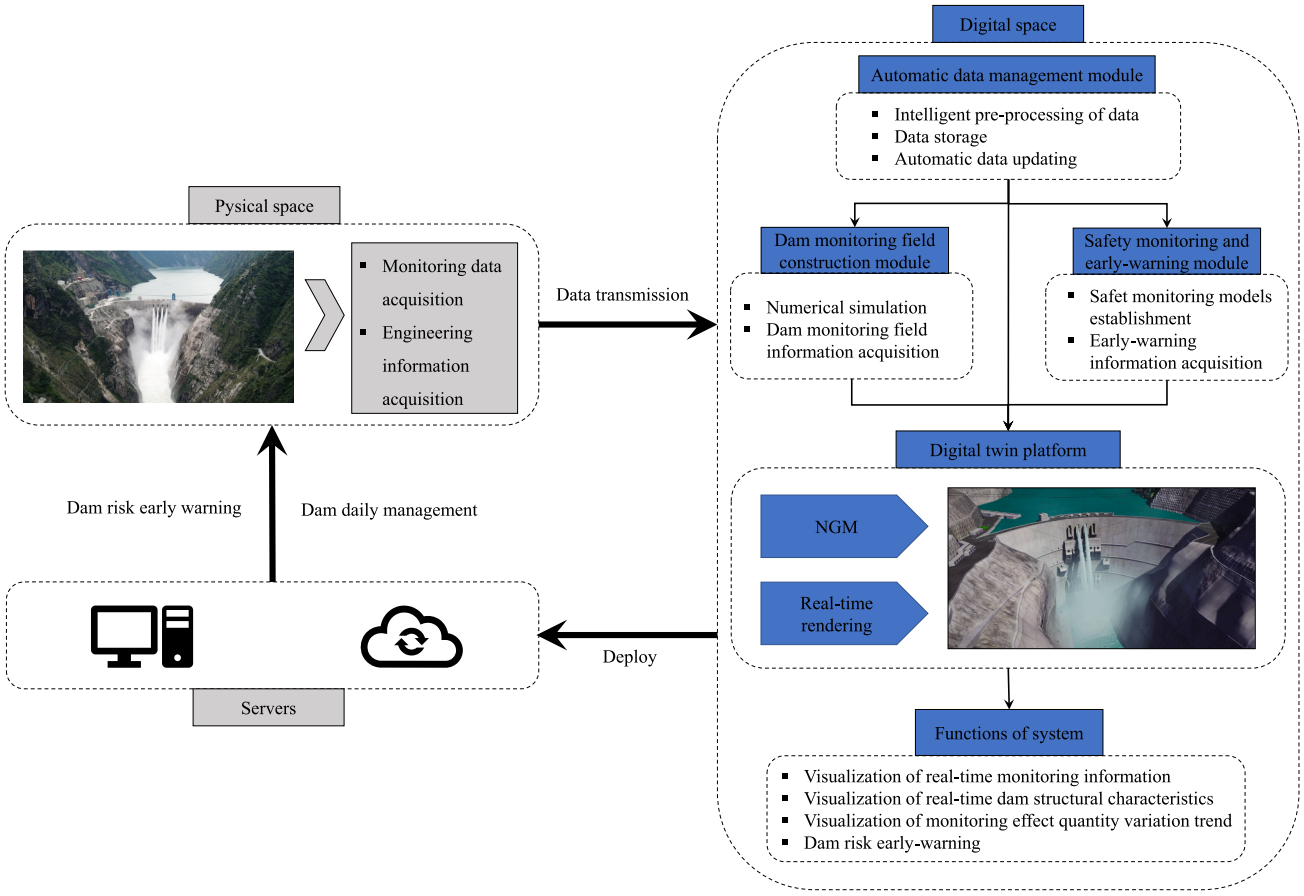


Fig. 1. Overall framework of the proposed system.

manage and analyze safety information of dams. Subsequently, the Electricity Generating Authority of Thailand (EGAT) initiated the Dam Safety Remote Monitoring System (DS-RMS), which focuses on acquiring information and evaluating dam safety condition remotely and timely (Suwatthikul et al., 2021). In comparison with KDSMS, DS-RMS possesses an additional rule-based expert system, which helps significantly evaluate dam state in future. Furthermore, some dam safety systems have also developed in China. For example, BIM model has been correlated with monitoring data in the dam safety management system (Zhou et al., 2023). Chang et al. (2018) introduced BIM model to effectively propose a platform which is able to integrate sensor information with BIM. The platform helps the facility manager to check monitoring data, which improves the intelligence of monitoring data management, analysis and early warning. Besides, project owners begin to pay more attention to the safety of basin systems, and Wireless Sensor Networks (WSNs) have been deployed in monitoring data collection (Rashid & Rehmani, 2016). Many problems occurred in the previous systems during their application. The traditional criteria for dam safety is not suitable because of its lack of real-time interactions and the complicated hydrological conditions. As a result, the SCU-DSMES system has been developed and applied in more than ten hydropower plants in China. A lot of effort has been spent on providing an entire support system platform for dam safety monitoring (Han et al., 2022).

In the process of visualization technological revolution hitting the dam safety monitoring system development, advanced video communication technology has been applied in dam condition remote viewing in China (Xiang et al., 2022). Apparently, the video consultation system plays a crucial role in dam safety real-time monitoring in Wangjia dam, Dahuofang power station and Ningxia water resources bureau. However, how to realize monitoring data and dam structural behaviour real-time display is still a problem to be solved. Consequently, Zhang et al. (2018) developed a platform to achieve dam information visualization and remote intelligent assistance for dam safety with the application of C# and GIS platform.

Meanwhile, the rapid growth of dam monitoring data volume is also a big challenge for achieving real-time analysis of dam structural characteristics. To improve data availability and reliability, various proven technologies, such as ASP.NET technology, Structural Query Language (SQL) and Geo-Information System (GIS) tools have been introduced in storage, pre-processing, and analysis of monitoring data (Li & Wang, 2017; Malik et al., 2020; Paul et al., 2020). Due to the outstanding readability and maintainability of SQL, the development efficiency of the system has been significantly improved.

To date, artificial intelligence (AI) technologies are changing the world, especially affecting the engineering management. In the domain of dam safety monitoring, the advanced technologies have been widely used in monitoring outlier detection (Zheng et al., 2022; Chen et al., 2024; Meng et al., 2024), safety monitoring model construction (Zheng et al., 2024; Li et al., 2021; Hu et al., 2020) and risk early warning (Chen et al., 2021). Furthermore, the concept of Digital Twins (DTs) was investigated quite intensively in recent years. A digital twin can be regarded as a virtual substitute of a physical system that enables real-time monitoring, analysis, and optimization of its performance (Juarez et al., 2021). The use of digital twins is attracting lots of attention in various fields, such as, manufacturing, energy, healthcare, transportation, and construction. The technology effectively reduces additional costs of physical experiments for testing different conditions in a real environment. There has been a turning point in developing dam safety monitoring system in accordance with the advent of DTs. The initiative aim of DTs is to digitise and informatize physical characteristics of a real dam, which provides a new solution to simulate and analyse complicated behaviours of dams (Kaewunruen et al., 2021; Lu

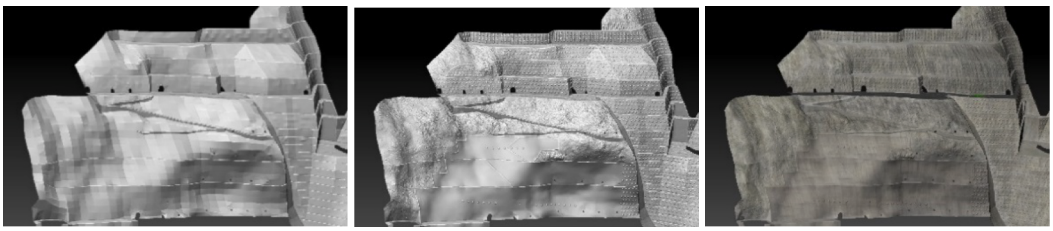


Fig. 2. Models of the right bank slop: (a) basic model, (b) high-fidelity model, (c) next-generation model.

et al., 2020). Boddupalli et al. (2019) presented a BIM-enabled platform to realize monitoring data visualization. Valinejadshoubi et al. (2019) have studied a new modular to query and visualize monitoring data. Zhu et al. (2023) integrated BIM technology and a multi-cloud dimension model to propose a DT-enabled solution for dam safety evaluation. Park & You (2023) presented a novel digital twin dam and watershed management platform with the application of GIS real-time data and AI technology.

In view of the evident advantages of AI and DTs in enhancing the real-time interaction and transparency of the dam safety monitoring system, these two technologies are suitable for intelligent system development. To the best of our knowledge, BIM has been utilized universally in DTs-based system. Nevertheless, it also has the shortcomings of insufficient modelling accuracy and limited universality. Although some efforts have been made to fine-grained modelling of engineering projects, the current three-dimensional digital model of engineering project still lacks of realism and immersion. Therefore, it's better to introduce novel modelling technology in three-dimensional model construction of dam. With the rapid adoption of next-generation modelling technique develop, it provides a possible way for designers to achieve higher quality and efficiency in 3D model production. The next-generation modelling technology (NGM) plays a crucial role in the undergone significant innovation of the gaming industry (Yang et al., 2023; Ryu & Hur, 2011). The next-generation modelling technology helps improve the highly detailed model intricacies and realistic material effects. To the best of our knowledge, although NGM has been used in modern game development, there are few applications of NGM in the three-dimensional model construction for engineering. Therefore, NGM can be introduced into DTs-based system of dam, which assumes the realistic meaning.

The main contributions of this paper are as follows. Firstly, the specific framework of the visualization system for dam safety monitoring with application of dam digital twin platform is proposed. Secondly, we clarify the approach of construct visualization model of the dam by applying next-generation modelling and real-time rendering technologies, which serve as a digital twin platform. C++ and Python programming languages, database technology and intelligent algorithms are used to process and analyse the real-time monitoring data, so as to reflect the real-time structural behaviours of the dam in the visualization window of the system. Thirdly, we apply this proposed system to a real super-high arch dam in China to test the suitability of the system. The practical adoption indicates that the system is able to exhibit the current safety state of the dam, at the same time, to realize real-time warning of the dam risk. The structure of the paper is organized as follows. Section 2 proposes the overall framework of the proposed system for dam safety monitoring. Section 3 presents the development methods of digital twin platform, automatic data management module, dam monitoring field construction module, and safety monitoring and early warning module. Section 4 shows the practical application of an arch dam in China. Section 5 concludes the whole paper and points out the future research directions.

2. The overall frame work of the proposed system

The overall framework of the proposed system is presented in Fig. 1. As depicted in Fig. 1, based on the establishment of the 3D model of the dam project, this model serves as a digital sandbox for displaying scene information. By seamlessly integrating the model with automatic data management, monitoring field construction, and safety monitoring and warning modules, monitoring data and deformation field results are directly displayed within the digital twin platform. This system enables intelligent preprocessing, storage, and updating of monitoring data, providing real-time visualization of the overall dam structural characteristics. Moreover, it also has functions of predicting the dam operational behaviour and risk warning information visualization. The proposed system collects and gets engineering information and monitoring data from sensors automatically. Finally, the system will be deployed on the servers or web platforms, enabling users to perform real-time dam risk analysis and daily management in time.

It can be noticed that two spaces existed simultaneously, the physical space and the digital space. The physical space primarily encompasses the actual dam structure, the surrounding environment, and the associated safety monitoring facilities. The digital space seems like a digital representation of the actual project. The digital twin of the dam is created by mapping the physical characteristics and operational data of the physical dam onto a digital model in real-time, resulting in a virtual "twin" of the dam. This process involves several key steps: Firstly, continuous monitoring of real-time data from the physical dam. Secondly, transmitting the collected data to the digital twin platform, where it is processed and analyzed. Thirdly, simulation and prediction, utilizing both real-time and historical data to accurately reflect and forecast the current state of the physical dam. Fourthly, decision support, enabling risk warnings for the dam and providing engineers with scientific insights for dam management, thereby optimizing operational and maintenance strategies to enhance safety and efficiency.

3. Design of the proposed system for dam safety monitoring

3.1. Digital twin platform construction based on NGM and real-time rendering technology

The digital twin platform serves as digital sand table for displaying the engineering scene effectively, including the dam body, dam foundation, reservoir basin, and surrounding mountainous environment. The platform incorporates information such as monitoring point locations, monitoring data, site information, and model information, enabling users to easily acquire the operational status of the dam. Therefore, the emphasis of developing the platform lies in constructing the visualization model with sufficient realism, and further simulating the actual engineering scenario through model rendering.

A common strategy used to establish the basic model is to collect the engineering data and drawings. This kind of model is characterized by its simplicity, low polygon count, and high operability, making it suitable for real-time rendering. However, the basic model suffers from a lack of detail and low precision, resulting in obvious discrepancies

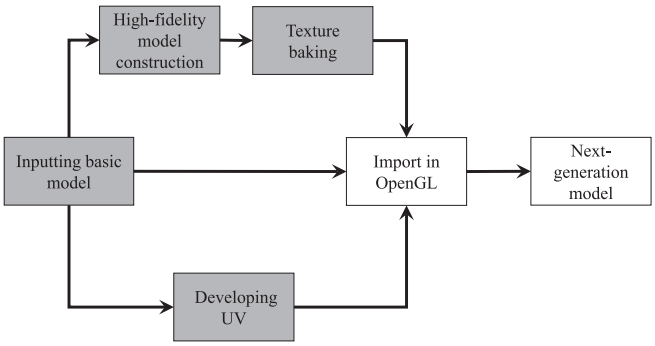


Fig. 3. The process of next-generation modelling.

between the rendered engineering scene and reality. To address this issue, a lot of effort should be spent on improving the mesh density and details. However, for the large-scale engineering structures like dams, high-fidelity models consist of thousands or even millions of polygons, it's not feasible and cost-effective to build high-fidelity models for all the structures.

To overcome this problem, the next-generation modelling (NGM) and real-time rendering technologies are introduced. After constructing a high-fidelity model, textures are baked from this model and mapped onto the basic model. This approach makes it possible to render the next-generation model with high-fidelity and high degree of realism in real-time. As illustrated in Fig. 2, the models of right bank slope of a certain arch dam are depicted. Wherein, Fig. 2(a) shows the basic model, Fig. 2(b) exhibits the high-fidelity model, and Fig. 2(c) describes the

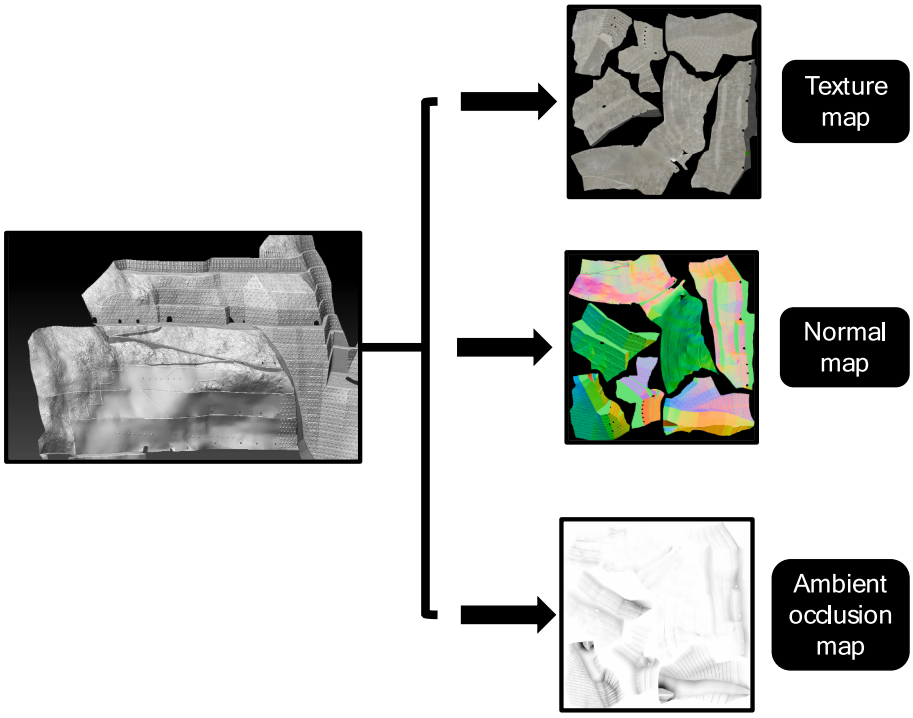


Fig. 4. The high-fidelity model and corresponding maps.

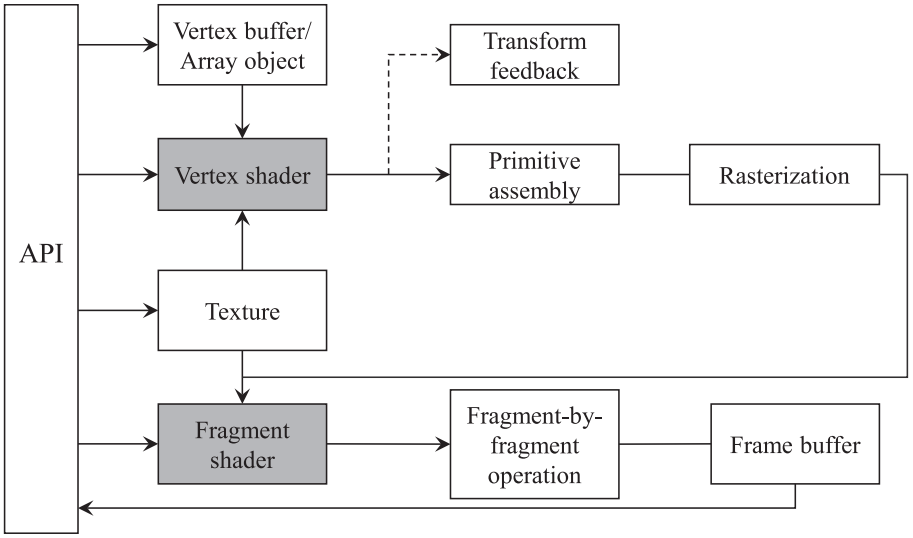


Fig. 5. Flow chart of programmable pipe of OpenGL.

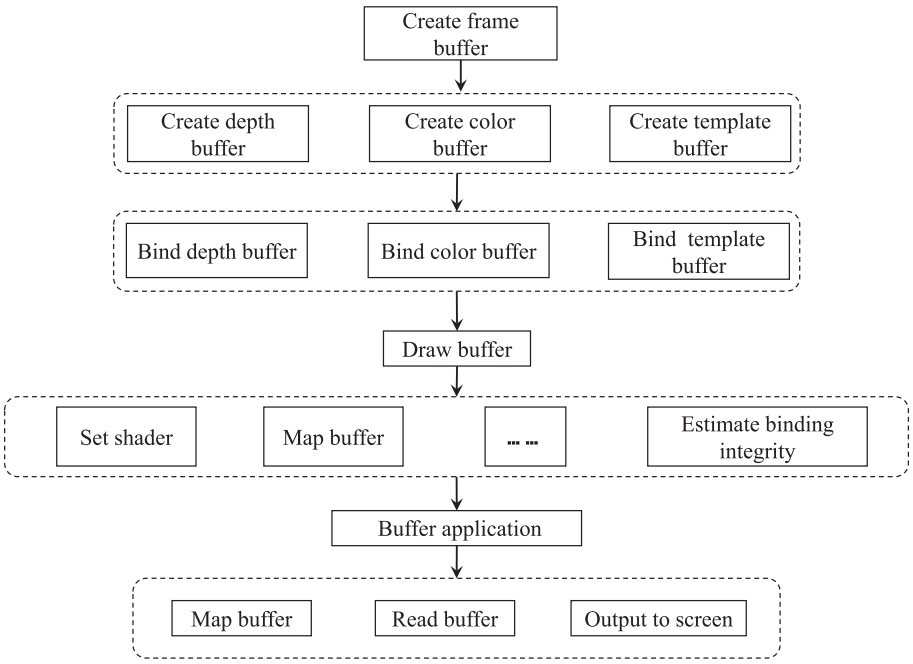
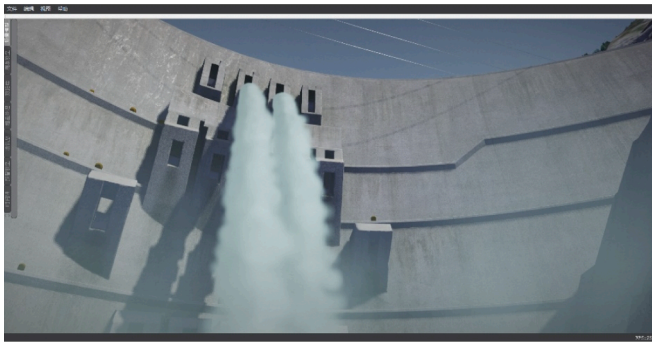


Fig. 6. Process of frame buffer technology.

(a)



(b)

Fig. 7. PBR rendered model of an arch dam: (a) the details of dam surface, (b) the dam when flood discharge.

next-generation model. The comparison clearly demonstrates that the next-generation model retains nearly all the details of the high-fidelity model, offering significantly more details than the basic model.

The specific steps of constructing next-generation model are as follows. After constructing the basic model based on engineering data and drawings, the next-generation model is created by using ZBrush, a commercial modeling software. The modelling process, as shown in Fig. 3, involves several steps, most of which are conducted within ZBrush (indicated by the gray boxes). In Fig. 3, "U" and "V" refer to the coordinates of texture maps that define the position of each point on the texture map. By unfolding the UV map of the basic model, the points on the texture can be linked to the 3D model, allowing for accurate placement of textures. Once the high-fidelity model is established, several common texture maps can be baked from it, such as, texture maps, normal maps, and ambient occlusion (AO) maps. Texture map primarily records the surface color of the model, representing the appearance. Normal map captures the direction of surface normal at each point on the surface of an object. By encoding these directions in the RGB color channels, normal maps are able to simulate the details of surface bumps and depressions. When lighting is applied, this map allows surfaces with fewer polygons to exhibit high-detail lighting and reflection effects. AO map enhances the realism of the model by simulating how ambient light is occluded or blocked in crevices or where objects intersect. It helps reduce light leaks and inaccurate shadowing, enhancing the realism of the model, particularly in shadowed areas.

Fig. 4 illustrates a high-fidelity model of the arch dam, alongside its corresponding texture map, normal map, and AO map baked from the high-fidelity model.

On the basis of the next-generation model, real-time rendering technique is employed to set up the environment, lighting, materials, and rendering parameters for the construction of the digital twin platform. Real-time rendering refers to the process of calculating and outputting graphical data in real time. In this study, the programmable rendering pipeline of OpenGL is utilized to achieve real-time rendering of the model. The programmable pipeline streamlines the rendering process by simplifying its logic, thus enhancing rendering efficiency. Moreover, it allows developers to implement specific algorithms and logic to render effects that are difficult to achieve with a fixed pipeline. The flowchart of the programmable rendering pipeline is depicted in Fig. 5, and the programmable stages are indicated by gray boxes. Utilizing this programmable pipeline, various rendering effects can be achieved, including the simulation of different weathers, seasons, and times of day in engineering scenarios. This helps make it possible to realize real-time visualization of flood discharge, accurately reflecting the actual engineering scene.

Fig. 8. HDR rendered model of the arch dam project.

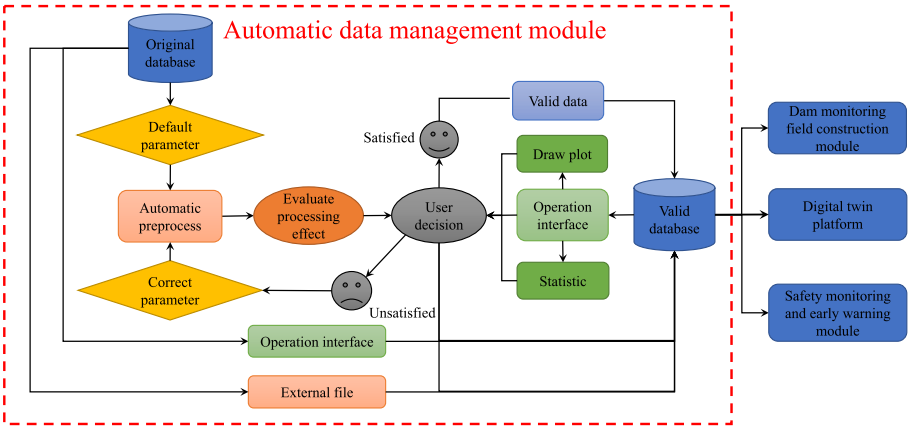


Fig. 9. Workflow of automatic data management module.

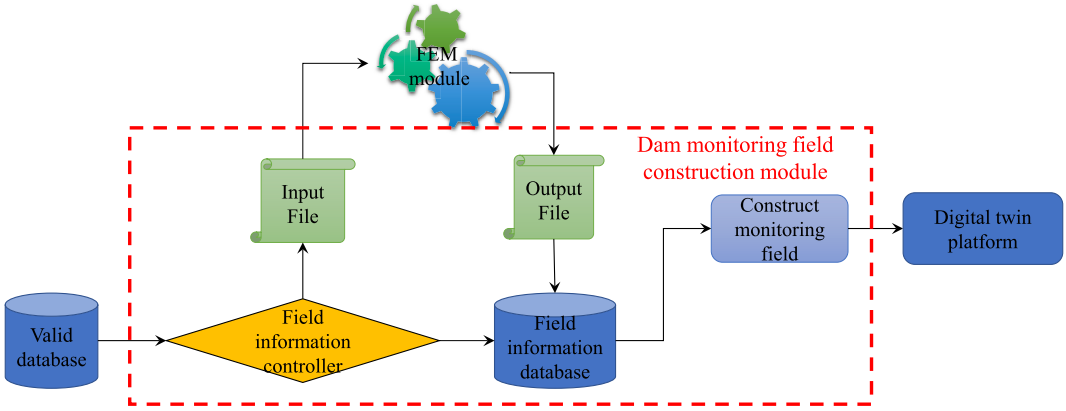


Fig. 10. The workflow of dam monitoring field construction module.

Moreover, the displayer achieves the desired rendering effects by continuously refreshing and displaying the pixels on the screen, which are derived from rendering data stored in the frame buffer. This data includes geometric information (vertex coordinates, texture coordinates, etc.) and textures processed through a series of rendering pipelines. OpenGL supports the use of multiple frame buffers simultaneously, allowing for various rendering effects through the creation of different frame buffers. Each frame buffer comprises three types of buffers: color buffer, depth buffer, and stencil buffer. The color buffer stores the color values of each fragment, with each color consisting of four channels: RGBA. The depth buffer holds the depth values of each fragment, representing the distance from the fragment to the viewpoint. The stencil buffer stores stencil values for each fragment, which are used for stencil testing to permit the discarding of specific fragments based on certain conditions. The workflow of frame buffer is illustrated in Fig. 6.

Among the real-time model rendering technologies, Physically Based

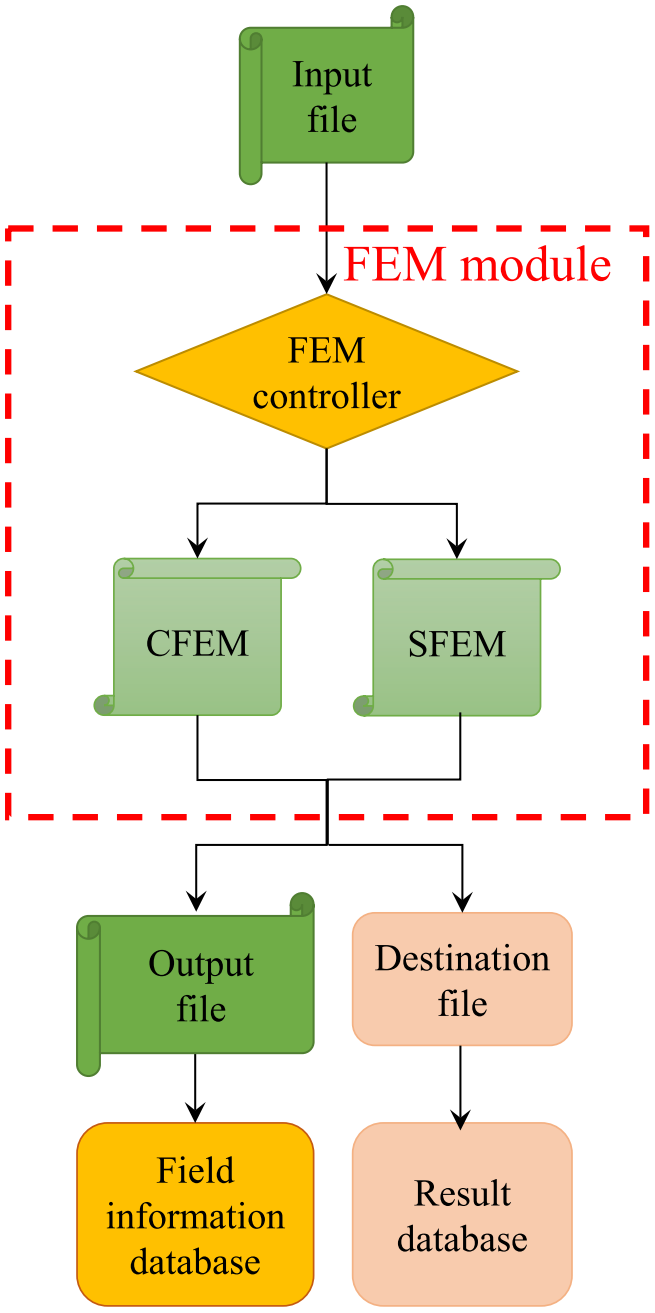


Fig. 11. The workflow of FEM analysis module.

Rendering (PBR) and High Dynamic Range (HDR) rendering techniques are important to increase the realism of the digital twin platform. PBR aims to simulate light in a manner consistent with physical laws. It involves defining surface materials based on physical parameters, which ensures that the appearance of the materials remains realistic under various lighting conditions. PBR (Physically Based Rendering) technology is primarily grounded in three theoretical foundations: the microfacet theory, the principle of energy conservation, and the Bidirectional Reflectance Distribution Function (BRDF). Based on these mathematical principles, the technique controls the response of specific surface points to light on a per-fragment basis. By configuring or adjusting commonly used maps in the PBR rendering pipeline at the texel level, material properties are defined to reflect the physical characteristics of real-world surfaces. The maps utilized include normal maps, metallic maps, roughness maps, ambient occlusion (AO) maps, and albedo (texture) maps, among others. Fig. 7 exhibits the PBR-rendered model of an arch dam. In Fig. 7(a), it is evident that the concrete surface of the dam reflects its real-world texture accurately after applying the PBR rendering technique. Fig. 7(b) illustrates the reflective effect on certain parts of the dam body caused by water stains, which appear after the floodgates are opened.

HDR is known as High Dynamic Range imaging, involves combining low dynamic range images captured with different exposure times into a single high dynamic range image. HDR (High Dynamic Range) rendering involves two primary steps. The first step is exposure control, which maps the high dynamic range image to the displayable range of (0,1) on the screen. The second step is to enhance particularly bright areas by introducing a bloom effect, creating the appearance of a radiant glow. This technique synthesizes the best details from each exposure to produce an image with a greater dynamic range and more image detail, thereby achieving more realistic visual effects that closely mimic the real-world environment. Fig. 8 presents the HDR rendered image of an arch dam. The rendering clearly enhances the contrast between light and dark areas, resulting in more pronounced depth and a heightened sense of the digital twin platform.

3.2. Development of automatic data management module

Relying on the constructed digital twin platform, a digital sandbox is able to demonstrate the functions of the system modules. Among all the modules, the automatic data management module is the most basic, which provides data backplane for the system. The automation of data collection, preprocessing, storage, updating, and visualization is achieved by the automatic data management module. Fig. 9 illustrates the workflow of the automatic data management module. As depicted in Fig. 9, original data can be read from original database, valid data can be identified by automatic preprocessing method. When users are dissatisfied with the results of data preprocessing, they can adjust the parameters of the automatic data preprocessing method through this

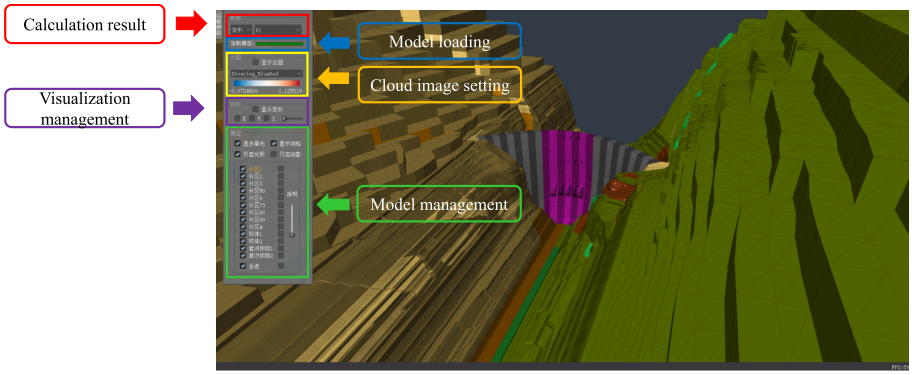


Fig. 12. The display interface of the result files.

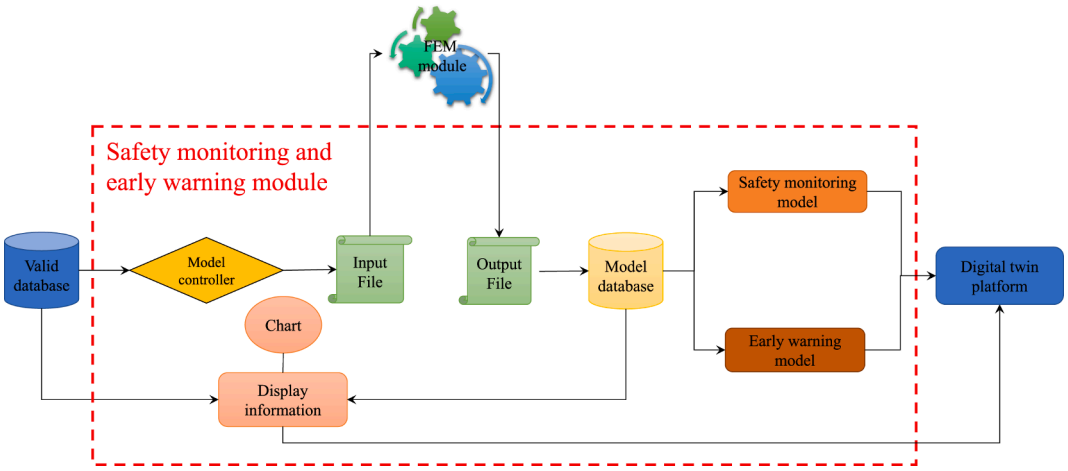


Fig. 13. The workflow of safety monitoring and early warning module.

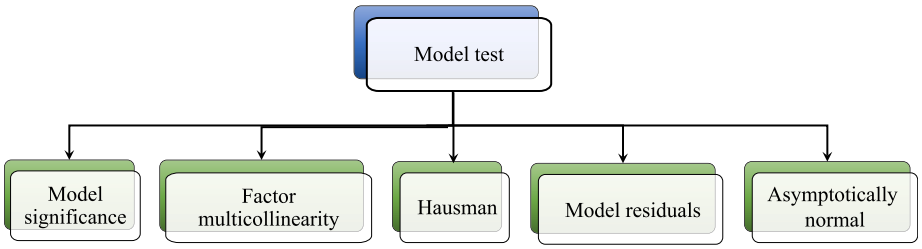


Fig. 14. The contents of various model validation method.

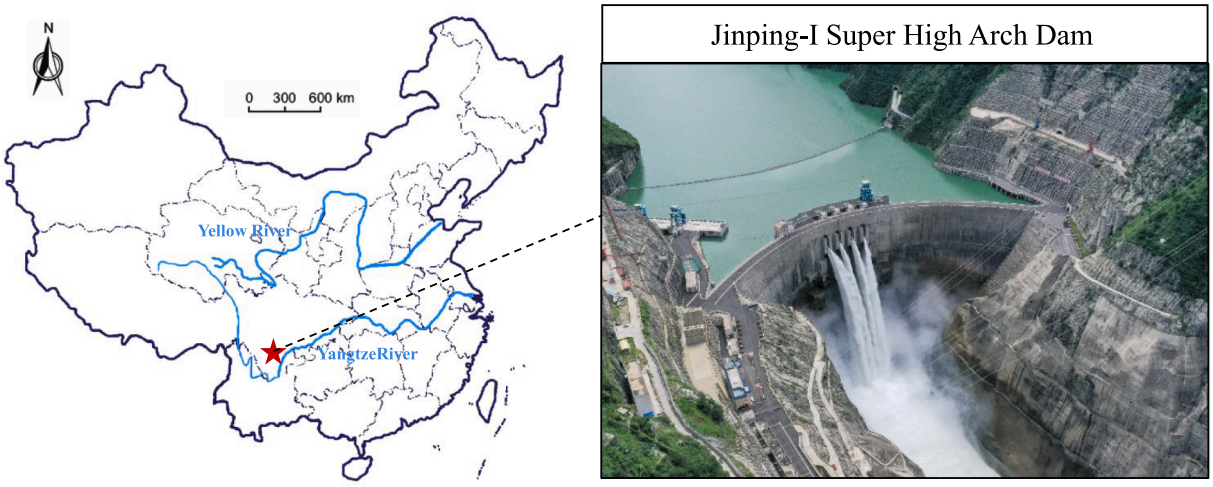


Fig. 15. A sketch map for the geographical location of Jinping-I super high arch dam and its photo.

module until satisfactory results are achieved. For valid data, the module supports data visualization through drawing data plot and statistical analysis. Additionally, valid data can be directly retrieved from the original database via external files or interface operations and finally stored valid data in the valid database. The automatic data management module interfaces with the digital twin platform, dam monitoring field construction module, and safety monitoring and early warning module. It can directly provide relevant information stored in the database to other modules for constructing monitoring fields and dam safety monitoring and early warning models. Additionally, the valid monitoring point data can be displayed on the digital twin platform according to the location of the monitoring points.

The automatic data management module consists of two databases: the original database and the valid database. The original database is utilized to store data automatically collected by the system, while the valid database contains the valid data after data preprocessing. This module adopts SQL Server, a relational database developed by Microsoft, to establish both the original and valid databases. A relational database is a type of database that organizes data using the relational model, which is generally easier to understand. The use of the standardized SQL language facilitates user interaction with the database, reducing the likelihood of data redundancy and inconsistency. By leveraging relational databases, users can efficiently collect and process the latest data from measurement points in real time.

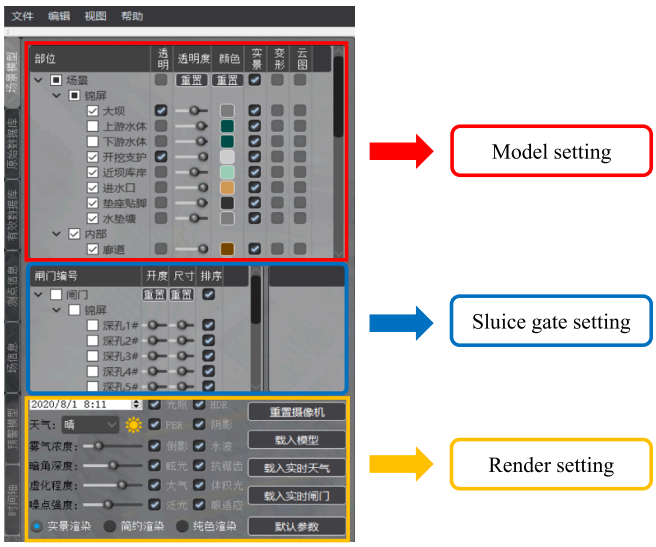


Fig. 16. The management window of digital twin platform.

3.3. Development of dam monitoring field construction module

The construction, information storage, and display of the dam monitoring field are primarily realized by the dam monitoring field construction module. This module utilizes a controller to import data from the valid database of the automatic data management module into the finite element analysis module. The output files are then automatically imported into the field information database for the dam monitoring field construction. Efforts have been made to link this model with the digital twin platform by integrating an interface for the dam monitoring field visualization. Fig. 10 illustrates the workflow of this module.

For the purpose of establishing an effective dam monitoring field, it is of great importance to develop a finite element analysis module capable of performing finite element numerical simulations. Fig. 11 presents the workflow of FEM analysis module. In Fig. 11, CFEM means Conventional Finite Element Method, SFEM means Stochastic Finite Element Method. They are two types of approaches used in the finite element analysis (FEM) primarily in engineering simulations. CFEM involves determining various physical quantities like stress, strain, temperature of the dam. SFEM is an extension of the conventional FEM that incorporates uncertainty into the model. SFEM aims to provide more realistic simulations in the presence of randomness and uncertainty. The input file generated by the dam monitoring field construction module is imported into the FEM analysis module for solving. This process generates an output file, along with a result file that is stored in the result database. The result file can be processed by post-processing software, allowing users to view the numerical simulation results directly. We have introduced the FEM analysis method proposed (Shao et al., 2023), and referred to the current open source software programs (Zaheer et al., 2020). The solver in the FEM analysis module is written in Python, and its source code is fully open, enabling facilitating development, maintenance, and learning.

Fig. 12 describes the display interface for the numerical simulation results. This interface allows users to load models manually and select specific calculation results. According to the selected results, users can adjust the visualization results of the finite element calculation model and the contour map of dam monitoring field.

3.4. Development of safety monitoring and early warning module

With the aim of applying the finite element analysis results to the safety monitoring and early warning model establishment, it is necessary to integrate the FEM analysis module into this module with an interface. After importing valid monitoring data from the valid database into this module, the generated processing files are further transferred to the finite element analysis module for computation. The computation results are then stored in the model database, allowing the establishment of various safety monitoring models. In the model database, the safety monitoring models include physical models, data-driven models and integrated models. The physical models are established by FEM based on the actual physical structure of the dam, these models simulate the dam mechanical behavior, deformation, seepage, and other factors. They typically consider the dam's geometry, material properties, external loads, and more. The data-driven models are built by analyzing large amounts of historical data using statistical and machine learning methods to assess dam safety and operational conditions. The integrated models combine physical models and data-driven models, performing comprehensive analysis using multi-source data to provide more accurate monitoring results. For the early warning models, there are three types: threshold-based early warning models, dynamic early warning models, multi-factor comprehensive early warning models. Threshold-based early warning models set threshold values for key parameters (such as water levels, displacement, stress, etc.). When these parameters exceed the safety range, an early warning is triggered. Dynamic early warning models analyze real-time monitoring data and use machine learning, fuzzy logic, and other methods to dynamically assess risks and issue early warnings. Multi-factor comprehensive early warning models assess the risks to the dam by considering the combined effects of multiple monitoring indicators (such as hydrological, meteorological,

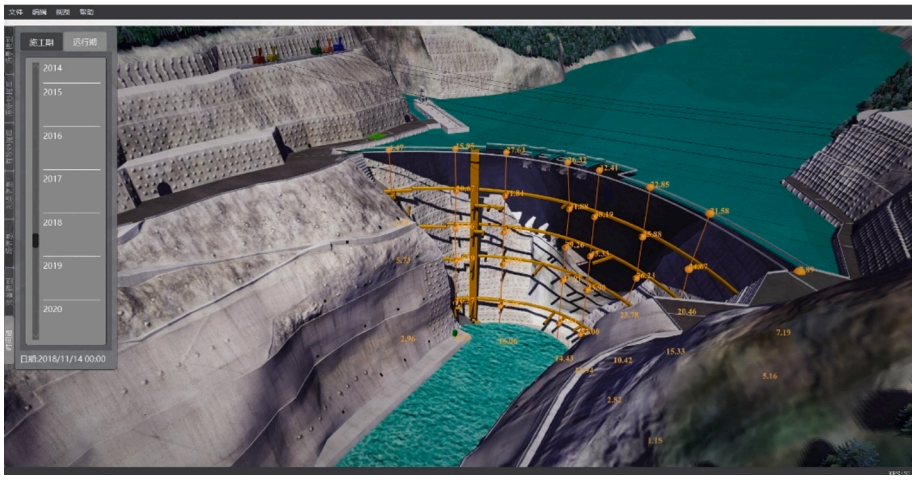
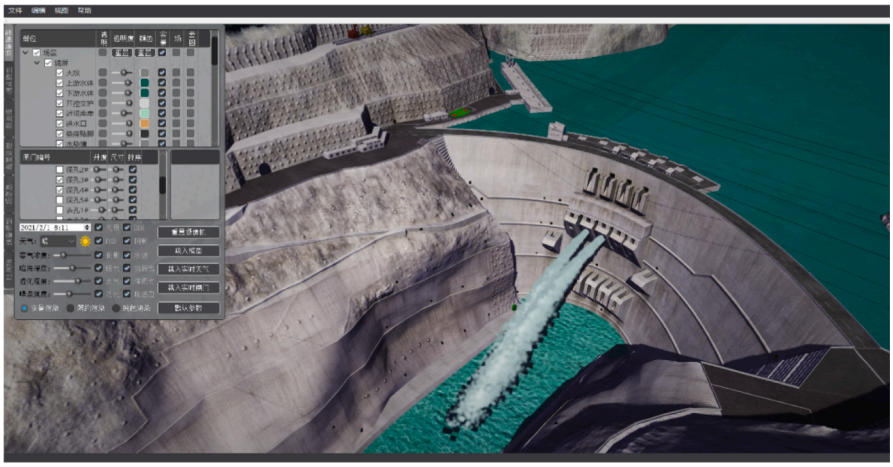
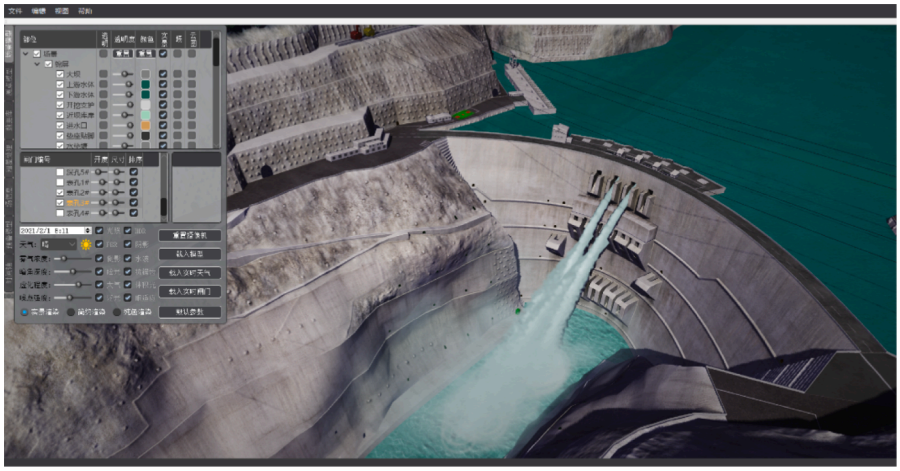


Fig. 17. The display of internal corridor of dam.



(a)

(b)



(c)

Fig. 18. Flood discharge scene of project: (a) when only deep hole gates 3# and 4# are open, (b) when both deep hole gates 3# and 4#, as well as surface hole gates 2# and 3# are open, (c) with increasing surface fog over time.

and geological data) and issue multidimensional warnings. Fig. 13 indicates the basic framework and analysis process of the safety monitoring and early warning module. The analysis results and data from different models can be directly displayed in the digital twin platform. To ensure the accuracy, effectiveness, and reliability of the models, the model validation methods are introduced into the system. With the aid of incorporating validation methods, the system is able to automatically verify the models against predefined criteria, ensuring that the models meet the required standards for application in safety monitoring and early warning processes. This approach not only enhances the credibility

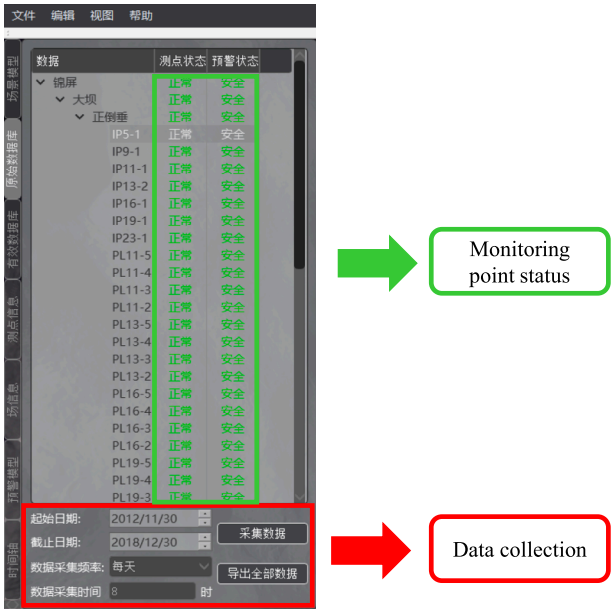


Fig. 19. The management window of original database.

of the models but also facilitates ongoing improvements and adjustments based on validation results.

As shown in Fig. 14, there are various model validation methods applied in the system. It outlines the key aspects assessed during validation, such as the accuracy of the prediction results, the consistency with the real monitoring data, and the robustness under varying conditions. Each of these aspects is crucial for confirming that the models are reliable and enable the accuracy of prediction and early warning. The validation process is designed to systematically test each component of the model, ensuring their overall correctness and practical implementation.

In the window of the proposed system, users can check the model validation and optimization results directly. Based on the results, users can select the safety monitoring models that exhibit lower structural risk and stronger generalization capabilities to establish.

4. Practical engineering application and analysis

In this section, the proposed system is tested in Jinping-I super high arch dam. The sketch map for the geographical location of the arch dam and its photo are shown in Fig. 15. It is a remarkable engineering achievement located on the Yalong River in Sichuan Province, China. As one of the tallest dams in the world, it stands at an impressive height of 305 m. The dam is a double-curvature arch dam, designed to efficiently handle the immense pressure of the reservoir water by transferring the load to the surrounding rock walls. Constructed primarily for hydropower generation, the dam supports the Jinping-I hydropower station, which has a total installed capacity of 3,600 MW. In addition to its energy production, the dam plays a significant role in flood control, irrigation, and water supply for the region.

4.1. Digital twin platform construction

Based on the next-generation models and real-time rendering effects, additional components such as a 3D visualization model, original database, valid database, monitoring information, safety monitoring and early warning models, and a timeline window are integrated to construct a comprehensive digital twin platform. Taking a real arch dam project as an example, the digital twin platform of the system is demonstrated. The constructed models for this project include: the arch dam, upstream and downstream water bodies, excavation supports, reservoir banks near the dam, intake structures, toe plates, corridors, underground caverns, power plants, roads, nearby mountain bodies, gates, vegetation, watershed terrain, and water bodies. Fig. 16 implies the window for the digital twin platform, through which model configuration, gate settings, and rendering options can be adjusted. The model setting includes the display of both internal and external views of various engineering structures, as well as the display settings for the nephograms of monitoring effects. The sluice gate setting includes the display of gate openings and gate sizes. The rendering setting includes the display of weather conditions, rendering effects, image quality and so on. Additionally, the rendering setting allows for the integration of real-time data, such as live weather information and the current operational status of sluice gates.

Table 1
The data sources and sensors involved.

Data source/ Monitoring content	Sensor type	Monitoring parameters	Application scenarios
Displacement monitoring	Displacement meter, inclinometers, GNSS sensors	Displacement, horizontal displacement, vertical displacement	Monitoring deformation of dam structures
Stress and strain monitoring	Strain gauges, stress sensors	Stress, strain	Analyzing stress within concrete and geotechnical materials
Seepage monitoring	Piezometers, pressure tubes, pore water pressure sensors	Pore water pressure, seepage flow	Monitoring seepage behavior within the dam
Water level monitoring	Water level sensors, pressure sensors	Reservoir water level, dam water level	Monitoring water levels upstream, downstream, and within the dam
Seepage velocity monitoring	Thermal flow sensors, electromagnetic flowmeters	Flow velocity	Detecting seepage velocity in the dam
Settlement monitoring	Settlement sensors, GNSS sensors	Vertical displacement (settlement)	Monitoring settlement of the dam and foundation
Temperature monitoring	Thermistors, fiber optic temperature sensors	Temperature	Monitoring temperature changes in dam concrete and the environment
Inclination monitoring	Tiltmeters, inclinometers	Inclination angles	Monitoring tilt of the dam and related structures
Vibration monitoring	Accelerometers, vibration sensors	Vibration acceleration, frequency, amplitude	Analyzing dynamic responses to earthquakes, blasting, or other vibrations
Crack monitoring	Crack meters, fiber Bragg grating sensors	Crack width, crack changes	Monitoring crack development in the dam and foundation
Meteorological monitoring	Temperature and humidity sensors, anemometers, rain gauges	Temperature, humidity, wind speed, rainfall	Monitoring environmental weather conditions affecting dam safety
Electrochemical corrosion monitoring	Potentiometers, electrochemical sensors	Potential, current	Analyzing corrosion of reinforcing steel
Turbidity monitoring	Turbidity meters, optical turbidity sensors	Turbidity	Detecting sediment content in dam leakage water

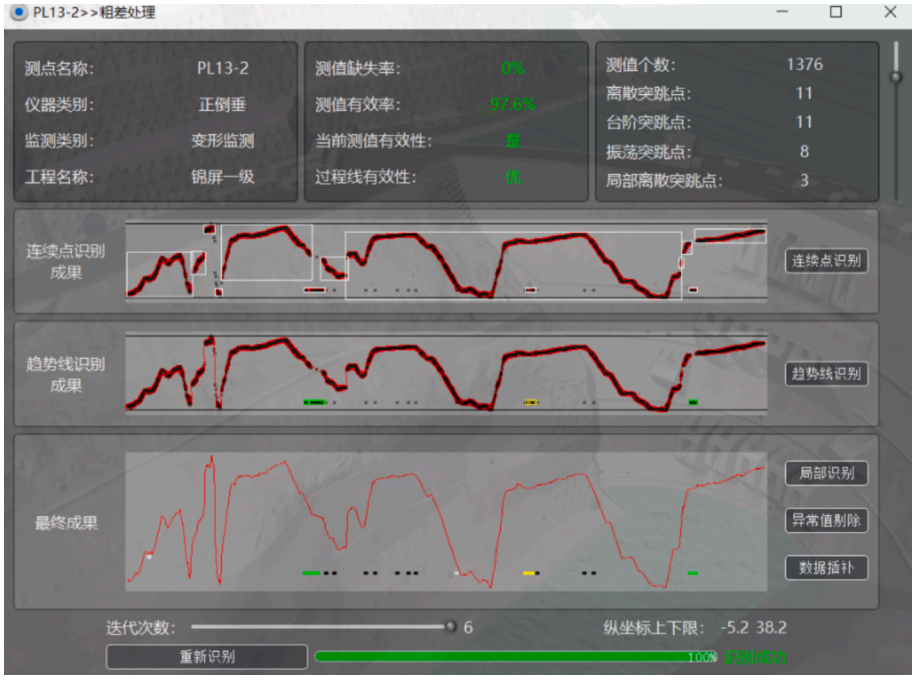


Fig. 20. The management window of data preprocessing.

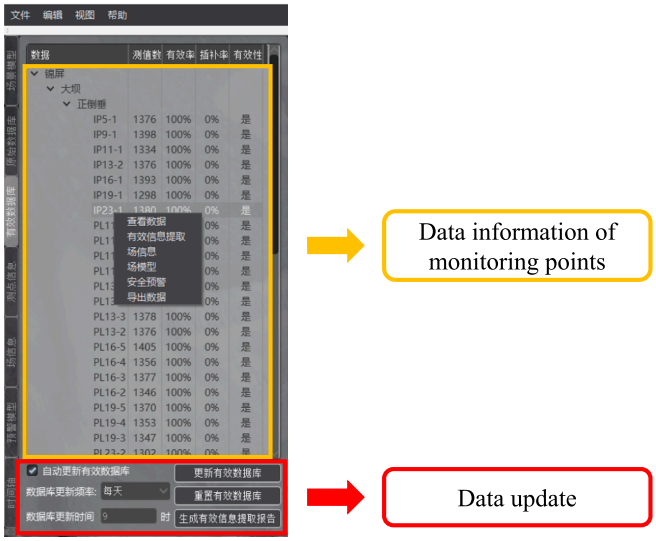


Fig. 21. The management window of valid database.

During configuring the model, users can manually select the engineering structures to display and adjust the transparency of the structures for detailed observation of the internal structures. Fig. 17 illustrates the display of internal corridors when the dam transparency is set to 90 %. By adjusting the gate openings and dimensions, engineering scenarios under different flood discharge conditions can be displayed on the platform. Fig. 18(a) shows the flood discharge scene when only deep hole gates 3# and 4# are open. Fig. 18(b) exhibits the scene when both deep hole gates 3# and 4#, as well as surface hole gates 2# and 3# are open, and Fig. 18(c) illustrates the discharge scene with increasing surface fog over time.

After selecting the rendering effects, the platform enables background blurring, vignette effects, noise, shadows, glare, atmospheric effects, bloom, water surface reflections, ripples, eye adaptation, HDR, and PBR rendering features. In addition, the system allows the loading of models, real-time weather conditions, real-time gate operations, and various rendering parameters to refresh and display real-time visualization engineering scenarios, reflecting the project status across different times, weather conditions, seasons, operational conditions, and environments. To accommodate different device capabilities, users can choose between simplified or realistic rendering modes. When only data visualization is needed, a solid color rendering mode is available.

4.2. *Display of the automatic data management module*

As described in Fig. 19, in the window of the original database, users can view the status of monitoring points and choose specific points for data processing. After setting the start and end dates for data collection, as well as the collection frequency and time, the collected data can be automatically imported into the original database. Moreover, users can manually export the collected data for further processing and analysis. Some data sources and sensors involved in this module are listed in Table 1 below. These sensors and data sources can provide comprehensive data support for the structural health and safe operation of dams. Additional specialized sensors may be included depending on the specific characteristics and monitoring requirements of the dam.

The management window of the automatic data management module is illustrated in Fig. 20. In the proposed system, we have applied the outlier detection methods that have been proposed (Shao et al., 2022) and then deployed the corresponding procedures in the system. After selecting the monitoring points, users can extract valid information through the data preprocessing window, as illustrated in Fig. 20. Once the window confirms that the data meets the prerequisites for preprocessing, the system will preprocess data and extract valid data. This management window displays the recognition results of data process line, on this basis, the outliers or gross errors that deviate from the data process line can be detected. Simultaneously, the statistical analysis and various types of outliers can be displayed on this window. And the validity of data from each monitoring instrument and monitoring category can be evaluated. Finally, the system automatically assesses the missing rate and the dispersion of missing data for each monitoring point.

The valid information can be utilized not only in the dam monitoring field construction, enabling real-time visualization of the dam overall

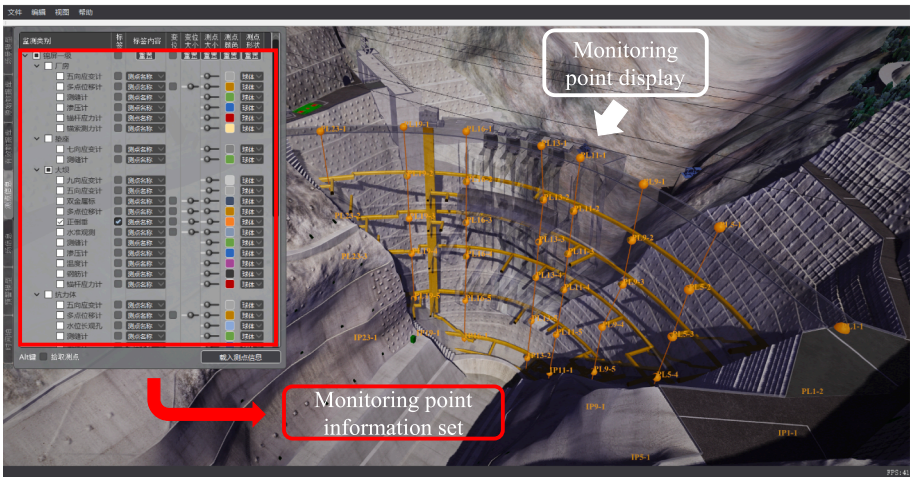


Fig. 22. The display of information and names of monitoring points.

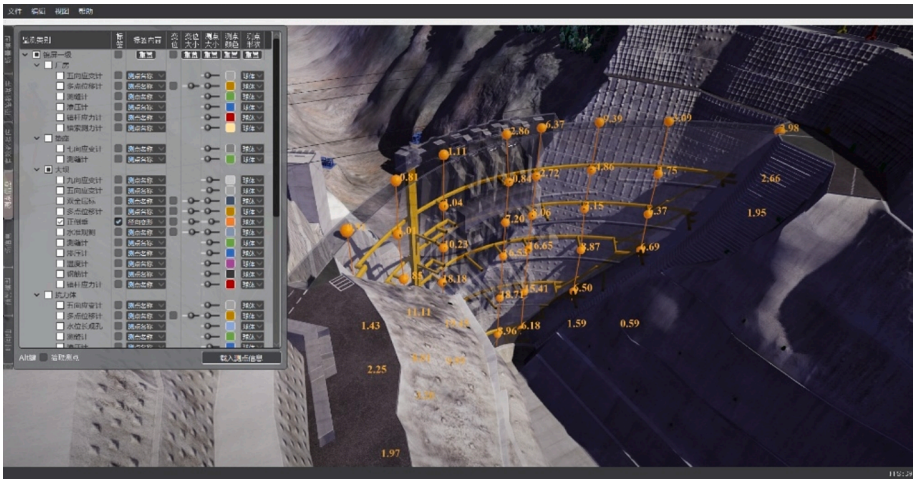


Fig. 23. The display of radial deformation of monitoring points.

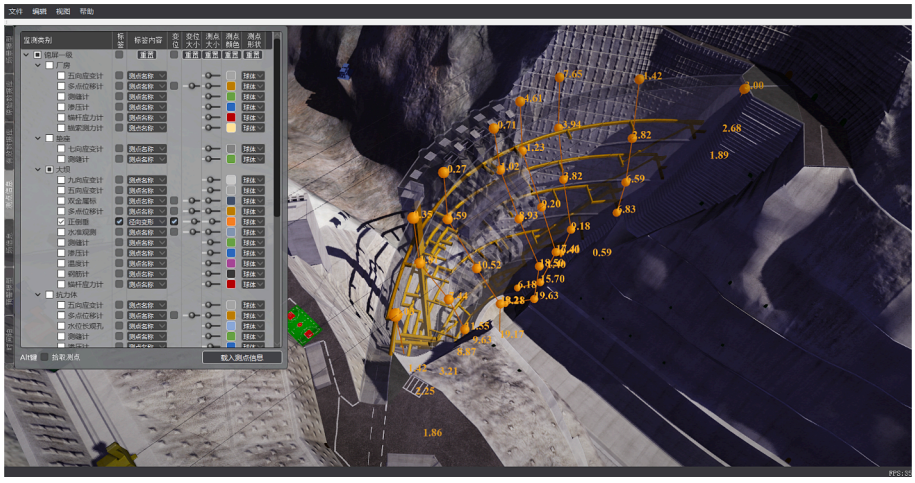


Fig. 24. The display of the deformation changes of monitoring points.

structural characteristics, but also the safety monitoring and early warning model establishment to predict and alert the dam safety status. Furthermore, users can customize and adjust various parameters based on the preprocessing results to achieve the desired data processing outcomes.

Through the valid database window shown in Fig. 21, by selecting the desired categories and information of monitoring points in the information window, users can directly view monitoring point

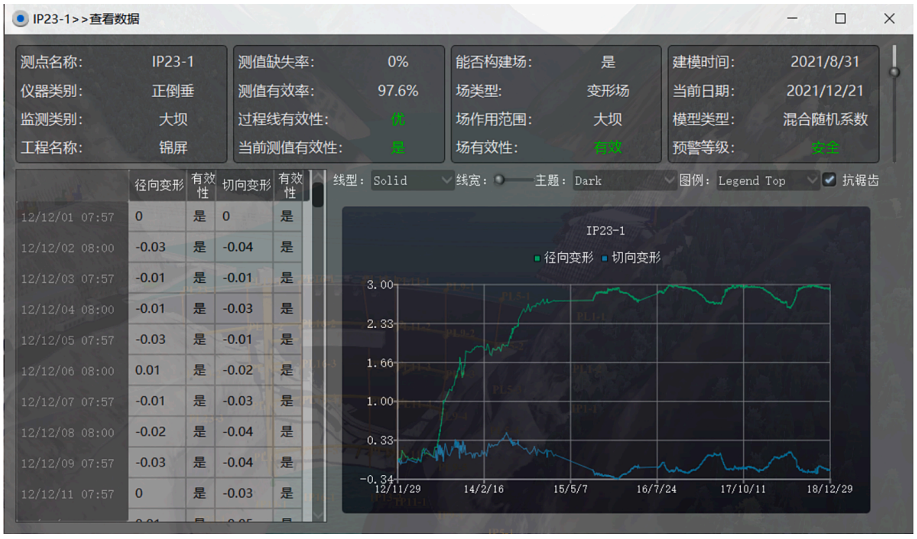


Fig. 25. The query window of valid data.

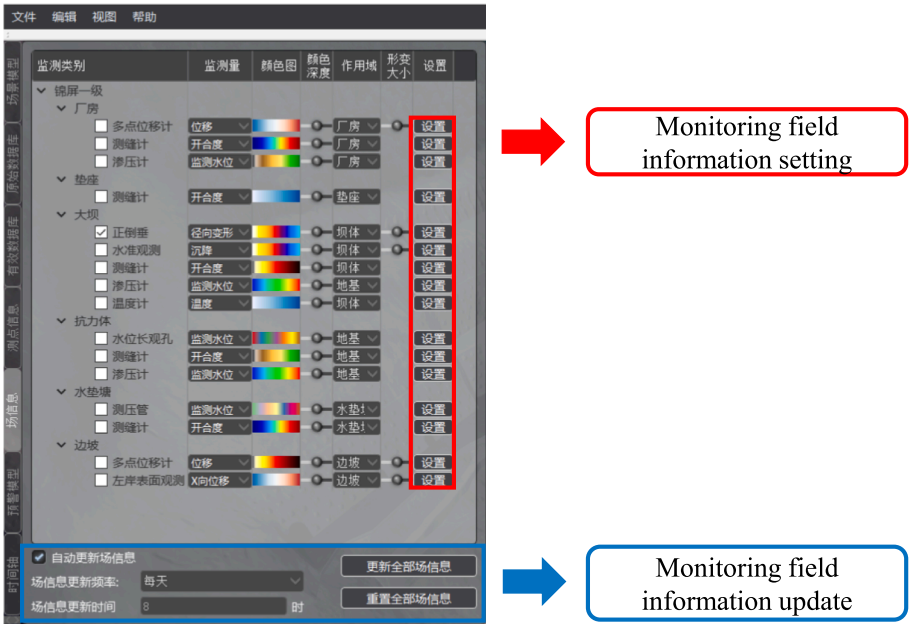


Fig. 26. The management window of dam monitoring field information.

information such as the number of monitoring values, efficiency, interpolation rate, and validity. The data can also be updated in this window. Besides, users can display the monitoring point information on the digital twin platform according to their locations. Fig. 22 illustrates the display of normal and inverted monitoring points and their names on the digital twin platform for the project. Fig. 23 shows the radial deformation values of the monitoring points as displayed on the platform.

Additionally, the system can dynamically display the monitoring information by setting the displacement magnitude, point size, color, and shape. Fig. 24 reflects the variation of radial deformation over time for the monitoring points on the arch dam. By dragging the slider in the timeline window, users can dynamically display the changes in radial deformation during both the construction and operating periods of the dam.

In order to view information for an individual point, the valid data display window (as shown in Fig. 25) allows users to view details such as the name of the monitoring point, instrument type, monitoring category, and data trend lines. Moreover, users can adjust the width or legends of the trend lines to analyze data variation trends. In addition, this query window can display the validity of the data and analysis results about whether it can be used for assessing dam behavior and establishing safety monitoring models. Real-time updating results on metrics such as monitoring data completeness, data availability, and risk levels are also presented in this window.

4.3. Display of the dam monitoring field construction module

Referring to Fig. 26, by selecting different monitoring metrics, users can construct various dam monitoring information fields. In addition, users can update or reset the dam monitoring field information in this system, such as the update frequency and time.

Taking the deformation monitoring field as an instance, Fig. 27 describes the deformation monitoring field construction window. By setting the deformation direction of the arch dam, the physical and mechanical parameters inversion and finite element numerical

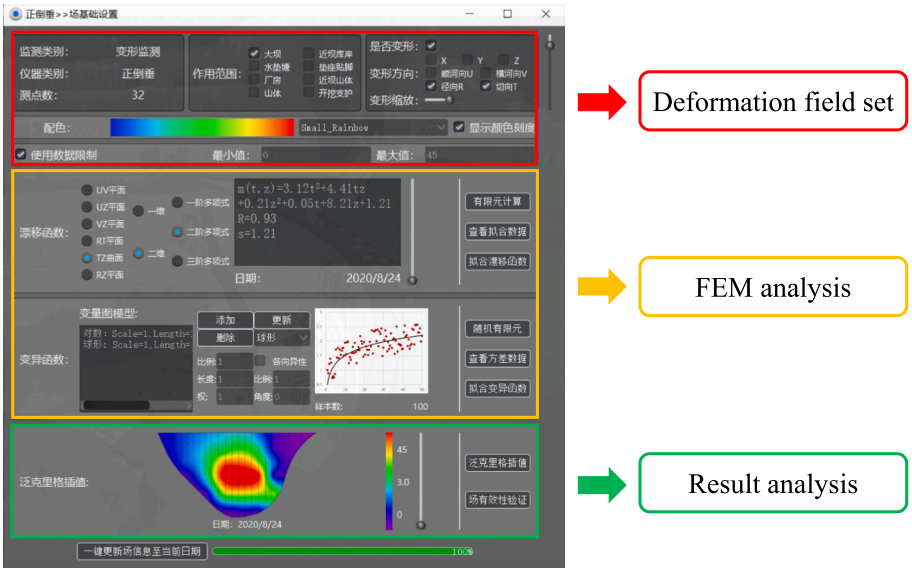
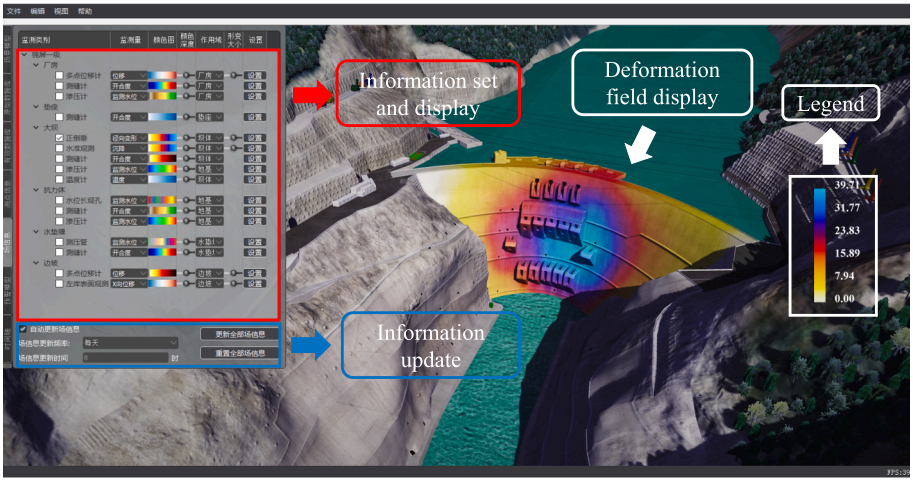
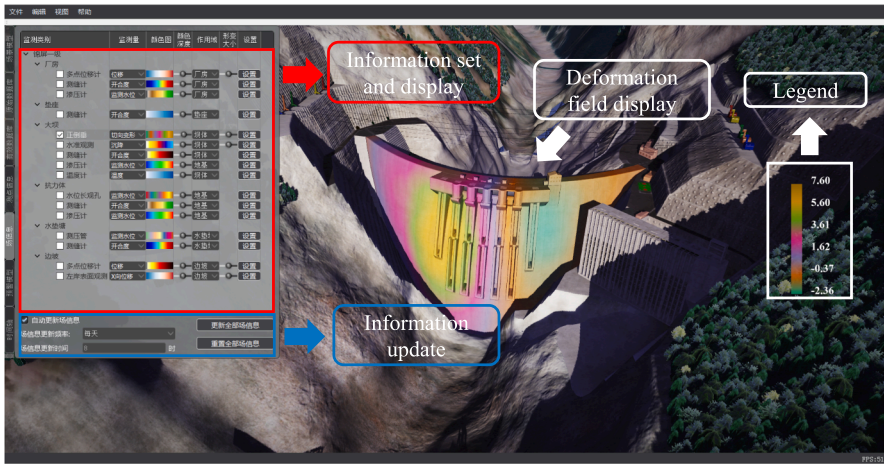


Fig. 27. The management window of dam monitoring field construction.



(a)



(b)

Fig. 28. The display of constructed deformation monitoring field: (a) the radial deformation field, (b) the tangential deformation field.

Fig. 29. The management window of model construction.

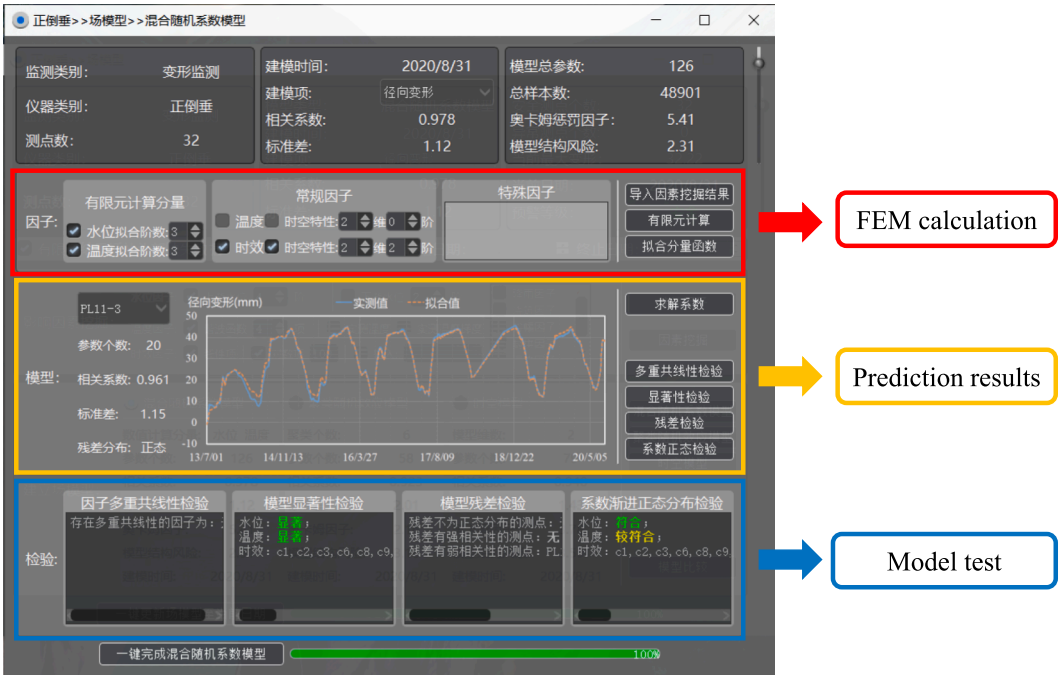


Fig. 30. The management window of model construction result.

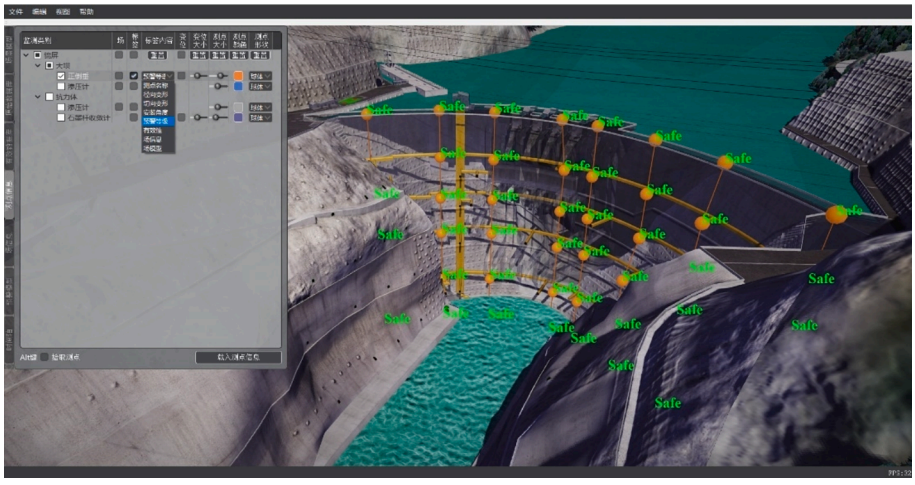


Fig. 31. The display of early warning results.

simulations could perform in the system. In this case, the deformation monitoring field of the arch dam can be constructed.

The construction results can be directly displayed on the digital twin platform. As shown in Fig. 28(a), the constructed dam radial deformation field, and Fig. 28(b) illustrates the constructed dam tangential deformation field. In the way of adjusting the color scheme, color depth, range of influence, and deformation magnitude, users can intuitively view the field information.

4.4. Display of the safety monitoring and early warning module

The management window for constructing the safety monitoring and early warning models is illustrated in Fig. 29. After updating the data, users can independently select the type of safety monitoring model to be developed and employ finite element numerical analysis methods to interpolate missing data. In addition to selecting the model type, the system allows users to select model factors and promptly displays information such as model validation results and the effectiveness of risk control measures of the model structure.

As illustrated in Fig. 30, through selecting the variety of safety monitoring models in the system, users can directly view the prediction and validation results in the management window. Furthermore, the early warning outcomes will display directly on the digital twin platform of the system as depicted in Fig. 31.

5. Conclusions

This study proposed a real-time NGM-enabled visualization system for dam safety monitoring with application of digital twin platform, which embeds computational programming, data processing, structural analysis, and health diagnosis domain knowledge. In the case of Jinping I super-high arch dam, the proposed system was tested to verify its specific functions. The results demonstrate that the developed system performs a more diverse set of functions as independent research and development without relying on existing software. The test results demonstrate the proposed system ultimately contributes to the overall structural safety status evaluation, and continuous supporting decision-making for dam safety monitoring. The main innovations and future work are the following as below.

(1) It presents a construction method of the digital twin platform based on NGM and real-time rendering technology. The construction result shows that the precision of 3D visualization model featuring a relative high precision, which is able to show engineering scenarios under different situations and helps to grasp the specific operation states of projects in real-time. This system incorporates digital twin platform with automatic data management module, dam monitoring field construction module and safety monitoring and early warning module for information visualization. In this case, the proposed system allows for dam monitoring data, structural behavior, safety monitoring result and early warning result visualization in real-time. With the application of the proposed system, it allows for more precise maintenance, helps in the early detection of potential problems, optimizes maintenance strategies, reduces the burden of manual inspections, and enhances operational efficiency.

(2) Meanwhile, the proposed system can integrate diverse data sources (e.g., on-site sensors, drone monitoring, meteorological data) into a unified database, facilitating information sharing among various departments in dam projects. Users can communicate and collaborate based on the same set of data, reducing information silos and improving overall efficiency. The digital twin platform in this system is not limited to safety monitoring, it can also simulate the surrounding environment. Through real-time monitoring and simulation, users can gain a better understanding of dam real-time operation state.

(3) Although the proposed system can realize continuous safety monitoring of dam characteristics and map the physical space into a virtual space, as verified in the application, there are still some limitations. For example, it can not predict how the dam might behave under extreme conditions such as earthquakes and floods in the virtual space, which still need further investigation. Besides, research on the database of emergency response plans in the system for preemptive actions to prevent disasters is also essential in the future work.

CRediT authorship contribution statement

Chenfei Shao: Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Visualization, Funding acquisition. **Sen Zheng:** Writing – original draft, Writing – review & editing, Visualization. **Yanxin Xu:** Investigation, Conceptualization. **Hao Gu:** Formal analysis, Methodology. **Xiangnan Qin:** Conceptualization, Software. **Yating Hu:** Conceptualization, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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